CHANAKYA UNIVERSITY, BENGALURU

INTEGRATED REPORT OF NEW PRODUCT DEVELOPMENT

Comprehensive New Product Development Report: SafeSip – Innovation in Women's Everyday Safety

SUMMATIVE ASSESSMENT

PRODUCT STRATEGY AND BRAND MANAGEMENT

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SUBJECT: Product Strategy and Brand Management

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We also thank all respondents who participated in our surveys, interviews and prototype feedback tests, as their contribution formed the foundation for our product decisions and market insights. Their inputs helped us refine our design, understand real-time user concerns and transform ideas into a meaningful solution.

Finally, we acknowledge the collaborative effort of our team members, whose continuous dedication, brainstorming and active involvement made the concept execution and report compilation successful. This project has been a valuable learning experience in research, product innovation and market application, and we proudly present this final integrated report with gratitude.

ABSTRACT

This report presents the integrated development and launch strategy of an innovative Smart Women Safety Water Bottle—a product emerging from a five-stage research-driven New Product Development (NPD) framework. The concept was derived from the rising concern regarding women's safety during daily commutes and the need for discreet, user-friendly protection tools that merge seamlessly into everyday life. Through consumer research, idea screening, concept evaluation, prototype testing and performance measurement, the Smart Safety Water Bottle was selected as the most viable solution offering dual benefits of hydration and personal security.

The product incorporates emergency SOS activation, GPS tracking, Bluetooth connectivity, LED alert indicators and hydration monitoring within a compact pastel-finished bottle, making it both functional and aesthetically appealing. Further, HOQ analysis, IRP planning, pricing models, and marketing strategy formulation guide the product from prototype to commercial launch. This final report consolidates findings from all stages—identifying the problem, validating user need, designing features, testing functionality and preparing market entry through a detailed promotion and launch framework. The product aspires to empower women, ensuring confidence, independence and safety through a simple, everyday object.

PREFACE

This report has been undertaken as part of the Summative Assessment based on New Product Development (NPD), comprising four systematic stages conducted over the course of the semester. Each stage of the project involved deep exploration, research, ideation, and analytical work to identify a real-world problem and translate it into a functional and market-ready solution.

The project progressed from recognizing women's safety concerns in everyday environments to generating innovative product ideas, validating consumer sentiments, analyzing technical feasibility, and finally converging towards a unique product concept — a Smart Safety Water Bottle designed for women. The report reflects every stage of this development journey, integrating problem identification, idea screening, concept testing, prototype development, HOQ & IRP insights, marketing strategy formulation, pricing decisions, and legal-launch planning.

This Stage-5 document consolidates and presents the complete transformation of a simple idea into a practical and socially relevant product. The focus lies on creating a safety mechanism that is discreet, accessible, and embedded into a daily-use object, ensuring that protection becomes effortless and always within reach. This preface marks the beginning of our final presentation — a culmination of research, teamwork, design thinking, strategic planning and creative execution.

PROJECT FOUNDATION AND STRATEGIC CONTEXT:

1. Project Overview and Name: SafeSip Innovation

The project is an academic and strategic exercise in **New Product Development** focusing on the "Women's Safety Products" category. The selected product, **SafeSip: The Smart Safety Water Bottle**, moves beyond traditional single-function safety devices (like pepper sprays or simple alarms) by adopting a **dual-utility strategy**. The name, "SafeSip," succinctly captures both core functionalities: "Safe" denotes the security aspect, and "Sip" refers to the primary, innocuous function of hydration.

This integrated approach is crucial in the context of modern personal security. Pure safety gadgets often suffer from low daily carry rates and high visibility, which can compromise the user in a threatening situation. SafeSip, however, is designed to be an **always-on, always-carried solution** that blends seamlessly into the user's daily routine. The project's success is defined not just by technological novelty, but by its ability to achieve **real-world usability and compliance** with the user's need for both security and discretion. The overall project serves as a comprehensive demonstration of applying structured NPD methodologies to a high-impact social problem.

2. Core Problem Addressed: Women's Safety and Vulnerability:

The fundamental market problem addressed is the persistent **fear and vulnerability** experienced by women, particularly in **unpredictable or isolated public spaces** such as while commuting, walking alone at night, or using ride-sharing services. Existing safety solutions often fail due to:

- **Low Accessibility:** Many high-tech solutions are expensive or require complex interaction.
- **Lack of Discreetness:** Visible safety devices (e.g., loud alarms) can escalate a situation or be targeted by an attacker.
- **Functionality Failure:** Reliance on unstable mobile apps or weak network connectivity.

SafeSip directly addresses these pain points by offering an invisible layer of protection. The goal is to provide women with a reliable, immediate, and easily accessible means of communication and deterrence, thereby transforming a commodity product (water bottle) into a **lifesaving utility**. The project's success is intrinsically linked to enhancing the user's sense of **autonomy and confidence** in managing their personal security risks. This emphasis on empowerment through discreet design is the social mandate of the product.

3. Ideation Process and Methodology:

The Ideation Process followed a disciplined, multi-stage approach rooted in **Design Thinking** principles to move from a broad problem statement to a refined product concept.

- **Open Brainstorming:** The group conducted open discussions, generating a wide array of potential solutions, prioritizing concepts that combined **high utility** with **discreet design**. Key initial considerations included: **Affordability** (critical for student segments), **Ease of Use** (essential under stress), **Technology Integration** (leveraging IoT and GPS), and **Real-Time Emergency Response** (necessity for immediate assistance).
- **Concept Clustering:** Ideas were grouped based on form factors (wearable, app-based, object-integrated).
- **Consensus and Shortlisting:** Through a structured internal debate and scoring mechanism, the team shortlisted four ideas that represented the most viable technological and market opportunities. This step was crucial for managing resource allocation and focus in the subsequent stages of development.

The methodology used a mix of **qualitative group discussion** (for creative idea generation) and preliminary **quantitative screening criteria** (for initial feasibility filtering), ensuring that the resulting ideas were not only innovative but also grounded in practical constraints and user needs, as documented in Stage 01.

4. Idea Shortlisting and Screening:

The project initially brainstormed multiple product ideas addressing women's safety.

Shortlisted Ideas

The team shortlisted four ideas:

1. **Smart Safety Bracelet**
   - **Features:** Wearable, SOS trigger, and vibration alert.
   - **Limitation:** Too easy for an attacker to notice or destroy.
2. **Self-Defence Smart Water Bottle (Chosen Finally)**
   - **Benefits:** Discreet, always carried, dual utility (hydration + safety), high user acceptability, and could integrate advanced electronics.
3. **AI-Powered Mobile Safety App**
   - **Features:** AI risk alerts and GPS route prediction.
   - **Limitation:** Requires network and handling the phone during fear.
4. **GPS Enabled Keychain Alarm**
   - **Features:** Easy to carry, loud siren.
   - **Limitation:** Looks like a safety device, so it's not discreet.

Idea Screening Criteria and Verdict:

The shortlisted ideas were evaluated using criteria such as **technical feasibility, market desirability, user acceptance, cost feasibility, discreetness, complexity, and daily carry comfort**.

Product Idea	Technical Feasibility	User Acceptance	Final Verdict
Smart Safety Bracelet	Medium	Medium	Rejected
Mobile Safety App	High	High	Supporting App, but not main product
GPS Keychain Alarm	Medium	Low	Rejected
Smart Safety Water Bottle	**High**	**Very High**	**Selected**

Product Innovation Charter (PIC)

The PIC outlines the background, focus, objectives, and constraints for the product:

Focus and Goals

- Develop products that provide **real-time protection and response**.
- Integrate technology and design to make safety devices **attractive, discreet, and easy to use**.

- Enhance **accessibility** with varied pricing for different income segments.
- Strengthen the brand reputation as a **socially responsible innovator**.

Objectives

- Find safety products that **align with women's daily lifestyles**.
- Ensure **technological reliability and user trust**.
- Create both high-tech and **affordable options**.
- Prepare **prototypes for field testing** in Phase 2.

Target Market

- **Demographic:** Women aged **15-45**.
- **Segments:** College students, working professionals, and homemakers.
- **Location:** Primarily urban and semi-urban areas.

Market Opportunity

The women's safety market in India is expanding due to:

- Increased safety concerns in public spaces.
- Government and NGO initiatives promoting women's protection.
- Rising disposable income for smart accessories.
- Social media awareness on self-defence and safety tools.

Constraints and Evaluation Metrics

- **Constraints:** Limited awareness in rural areas, cost of integrating advanced sensors, privacy concerns with location tracking, and the need for strong battery life and reliability.
- **Evaluation Metrics:** User acceptance and satisfaction, cost-efficiency and durability, technology reliability and alert response time, and aesthetic appeal and market readiness.

5. Market Research and Problem Definition

The market research used a combination of **quantitative (Structured Survey) and qualitative (In-depth Interviews) research methods** to understand safety concerns, hydration habits, and product acceptance.

Research Design

- **Total Sample Size:** 30 female respondents.
- **Sample Profile:** Included college students, working women, homemakers, and women who frequently commute, across age groups 18-45.
- **Data Collection:** Questionnaire (Google Forms/printed) and one-on-one/informal interviews.

Problem Definition

The core problem is the increasing concern among women about personal safety, especially during commuting, travel, and late-night movement. Traditional safety tools are often non-discreet, difficult to access quickly, or easily forgotten.

The research aimed to answer three main questions:

1. Do women need a **discreet, safety-integrated everyday product**?
2. What **features** would they value most?
3. How **acceptable** is the concept of a smart water bottle with built-in safety functions?

6. Product Concept and Protocol

Product Concept (The Smart Safety Water Bottle)

The product is a **dual-purpose hydration + hidden safety device**. The concept statement is: **"A stylish everyday water bottle with built-in safety technology to protect women discreetly and instantly"**.

Core Features

- Hidden **SOS button**
- **Real-time GPS location sharing**
- **Bluetooth-enabled app triggering**
- **LED notification safety ring**
- **24-hour rechargeable battery**
- Hydration tracker
- Anti-slip grip & pastel feminine design
- Silent vibration mode for discreet emergencies

Category	Requirements	User Needs Fulfilled
Product Specs [76]	**Capacity:** 650-750ml, **Weight:** Lightweight < 380g, **Durability:** IP64 water resistance	Feeling safe while travelling, quick emergency alert system
Safety Activation[81]	**Activation Speed:** SOS activation < 2 sec, **Alert:**120dB alert option	Quick one-hand activation
Performance[85]	**GPS:** Accuracy < 10 meters, **Battery:** Full charge = 24-36 hours, **Bluetooth:** Range = 10-12m	Device that blends into normal daily objects, no embarrassment/suspicion

7. Pricing, Revenue, and Variants

Pricing Model

The product will be offered in three variants with different features and price points:

Variant	Features	Price Range (INR)
Basic	SOS button, app integration, GPS	₹1,499 - ₹1,999
Premium	Built-in GPS, loud alarm, fall detection, LED alert	₹2,499 - ₹2,999
Sports/Outdoor	Rugged built quality, larger battery	₹3,200 - ₹3,800

Revenue Model

The revenue streams include:

- **Product Sales:** Direct sale of bottles across online platforms and retail stores.
- **Subscription Model:** Premium tracking/cloud features at **₹49–₹79/month**, and advanced analytics.

- **Accessory Sales:** Protective sleeves, replacement lids, charging cables.
- **Corporate/Institutional Partnerships:** Bulk deals with colleges, gyms, women's hostels, and corporate offices.

8. House of Quality (HOQ):

The House of Quality table connects **Customer Needs (WHATs)** with **Technical Features (HOWs)** to analyze their relationships.

Customer Needs (WHATs)	SOS Module	GPS Accuracy	Bluetooth	LED Alerts	Battery Capacity
Feel safer while alone	Strong (●)	Strong (●)	Medium (○)	Weak (○)	Strong (●)
Quick help alert	Strong (●)	Strong (●)	Medium (○)	Weak (○)	Medium (○)
Discreet Feminine Design	Weak (○)	Weak (○)	Strong (●)	Strong (●)	Medium (○)
Must be easy to carry	Medium (○)	Weak (○)	Strong (●)	Medium (○)	Weak (○)
Battery must last long	Weak (○)	Medium (○)	Weak (○)	Weak (○)	Strong (●)

Legend: ● = Strong relationship, ○ = Medium relationship, ∘ = Weak relationship

9. Launch Plan and Limitations

Launch Plan (5 Phases):

- **Phase 1: Prototype Development (Month 1–2):** Finalise CAD design, 3D print outer shell, test PCB/battery integration, and conduct internal safety stress-testing/drop tests.
- **Phase 2: Controlled User Testing (Month 3):** Engage 20–30 female volunteers to evaluate grip, SOS responsiveness, and weight comfort, then revise based on feedback.
- **Phase 3: Pilot Launch (Month 4–5):** Launch in college campuses and female workplaces with discounted early-access pricing, collecting behavioural usage analytics through the app.
- **Phase 4: Full-Scale Launch (Month 6):** Availability on Amazon, Flipkart, and retail outlets, with influencer marketing and tie-ups with gyms, cabs, and universities.
- **Phase 5: Expansion & Upgrade (Month 7 onwards):** Introduce subscription-based tracking plans and a rugged outdoor edition.

Project Limitations:

- **GPS performance** depends on network availability.
- **Battery life** may vary with alert usage.
- **SOS button** must avoid accidental activation.
- **User trust-building** requires market education.
- **Waterproofing electronics** is challenging.
- **Production costs** fluctuate due to chip prices.